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https://www.usenix.org/conference/soups2026/presentation/ramat

This paper is included in the Proceedings of the
Twenty-Second Symposium on Usable Privacy and Security.

August 24–26, 2026 • Hannover, Germany

ISBN 978-1-939133-59-5

Open access to the Proceedings of the
Twenty-Second Symposium on Usable Privacy and Security
is sponsored by USENIX.

Passkeys in the Wild: A Systematic Study of FIDO2 User Experience Consistency Across Websites

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Abstract

Passkeys have emerged as a phishing-resistant alternative to passwords, yet little is known about how consistently they are implemented across the web. Prior usability research suggests that cross-site inconsistencies increase cognitive load and hinder adoption, as users carry expectations from one website to another. To address this knowledge gap, we conducted a systematic analysis of passkey user experiences across 111 websites using a comparative framework of 28 factors. Cluster analysis reveals that core passkey support is converging, but supporting features remain uneven. Future research can use our framework for longitudinal comparisons and studies targeting specific factors of interest.

1 Introduction

User authentication is essential for safe and secure online experiences. Historically, websites have relied on passwords for user authentication [22]. However, passwords are often difficult to remember and have fundamental security weaknesses [10]. To address these issues, the Fast Identity Online (FIDO) Alliance was established in 2012 [14] to develop and promote authentication standards designed to reduce the world's dependence on passwords [14]. One such alternative is the passkey [15], an authentication credential that allows users to log in to online services.

Passkeys offer a simpler, more intuitive, and more efficient way to authenticate than passwords [52]. They are designed to be more convenient for users by removing the need to remember and type passwords. Passkeys enhance online security by using public-key cryptography instead of passwords [42]. To improve the usability of passkeys, the FIDO Alliance has developed design guidelines for their implementation [13]. As more websites implement these guidelines, the passkey user experience should become more consistent. This matters because, according to Jakob's Law of Internet User Experience, users expect websites to behave like those they have already used [38]. Similarly, Lyastani et al. [32] emphasized that consistency helps websites align with users' mental models and avoid increasing cognitive load and friction by forcing them to learn something new.

Although several passkey implementation guidelines exist, prior research has not examined whether the passkey user experience is consistent across websites. To address this knowledge gap, we conducted a comparative analysis of passkey implementations across 111 websites to gain insights into deployment practices, assess the consistency of the passkey user experience, and identify problematic design patterns. We structured our evaluations around the following research question: *How consistent is the passkey user experience across the websites in our dataset?*

To answer our research question, we selected websites from `passkeys.directory` [1] and mapped user journeys for passkeys by exploring and documenting all possible user navigation paths for setting up, using, and deleting passkeys. We then established comparison factors to analyze passkey implementation practices across these sites and assess how each website applied them. Next, we analyzed the collected data using Euclidean distance and hierarchical clustering to evaluate the consistency of the passkey user experience.

We chose a comparative evaluation approach instead of direct user studies to analyze real-world passkey implementations at scale and assess the current state of the ecosystem. Our analysis identified 28 comparison factors covering different stages of the user journey, including discovery, setup, usage, and deletion. Each factor represents a specific design or implementation decision, such as how passkeys are promoted, prioritized over passwords, explained to users, or integrated into cross-device flows.

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USENIX Symposium on Usable Privacy and Security (SOUPS) 2026.
August 23–26, 2026, Hannover, Germany.

Conducting comprehensive user studies on all these variables would require an impractical number of conditions and participants, making full coverage infeasible. Instead, we evaluated each website by categorizing key moments in its passkey implementation as a match, partial match, or non-match based on specific comparison criteria. This structured comparison approach allows future research to focus on individual factors for detailed user studies, rather than trying to assess the entire passkey experience at once.

This study makes three main contributions. First, it introduces a comparison framework of 28 passkey features, which serves as a basis for systematically evaluating passkey implementation practices across websites. Second, we provide an empirical analysis of the consistency of passkey user experiences, highlighting common design inconsistencies and problematic patterns. Finally, we provide insights into potential future research directions to enhance the passkey experience.

2 Background and Related Work

A passkey is a discoverable FIDO2 credential that enables passwordless authentication using public key cryptography [15]. Passkeys may be device-bound, such as when stored on a hardware security key (HSK), or synchronized across multiple devices through credential managers, such as iCloud Keychain (Apple Passwords) [3] or Google Password Manager [20]). The FIDO Alliance uses the term "passkeys" to refer to both synced and device-bound credentials [40].

During registration, the client creates a key pair; the private key remains on the authenticator while the relying party stores the public key. During login, the relying party issues a challenge that is signed by the private key after local user verification (biometrics or PIN), and the signature is verified using the stored public key. If the verification succeeds, the user gains access [34].

2.1 Comparative Evaluation of Authentication Systems

Passwords are prevalent, and authentication failures are often systemic rather than user-driven [6, 43]. Studies on adaptive and layered authentication mechanisms, such as risk-based authentication (RBA) and multi-factor authentication (MFA), show that implementations are frequently inconsistent and weakened by fallback and recovery flows [17, 18, 21, 26, 31, 51].

Frameworks help compare authentication systems. Some classify authentication schemes generally [5, 39], while others focus on graphical authentication [4, 46] or biometrics [23, 44]. Others support authenticator selection such as Kontun [50], NIST SP 800-63-4 [49], or ACCESS [53].

Researchers have also developed frameworks for evaluating authentication schemes and real-world implementations [6, 43]. These studies provide useful ways to compare authentication mechanisms, but they focus primarily on passwords or authentication practices more broadly. More closely related to our work, Lyastani et al. systematically analyzed the consistency of two-factor authentication (2FA) user journeys across top-ranked websites [32]. Their comparative study inspired our work. Our methodologies are similar, and we categorize user journeys in almost the same way. The main difference is that we derive our comparison factors through a top-down analysis guided by the FIDO Alliance Design Guidelines.

2.2 Usability and Adoption

A growing body of research has examined the usability and adoption of FIDO2 and passwordless authentication. Controlled experiments and real-world deployments suggest that security keys and platform authenticators are generally perceived as more usable and secure than passwords [7, 12, 24, 25, 27, 28, 33, 52]. However, prior work highlights persistent challenges, including biometrics, fallback mechanisms, authentication options, device loss, multi-device use, and mental models [25, 27, 33, 37, 52].

Research on user perceptions shows that adoption is driven less by technical understanding and security, and more by perceived usability, self-efficacy, and response cost [2]. Enterprise studies further emphasize lifecycle complexity, legacy integration, regulatory constraints, and inconsistent platform support as major barriers [24, 25, 28, 29].

Importantly, most prior work relies on controlled studies, mock systems, or single-organization deployments, offering limited insight into how passkeys are implemented and communicated across diverse real-world services. Our work complements these studies by providing an ecosystem-level analysis of passkey user experience consistency in the wild.

2.3 Design Guidelines

The FIDO Alliance produced a set of Design Guidelines [13] to ensure passkeys are implemented consistently across the Internet. Additional guidelines have been produced by Google [19], Microsoft [35], and Authsignal [16]. In our work, we focus on the FIDO Alliance Design Guidelines as a stable and widely supported baseline.

The FIDO guidelines have three categories of design patterns: required, optional, and security-key. Required patterns enable creation, management, and authentication. The optional patterns offer enhanced functionality that users may reasonably expect. Security-key patterns support FIDO2 hardware security keys.

By focusing on the FIDO guidelines, our comparisons only include advice reasonably available to all implementers. We show how these are applied in practice, revealing gaps between guidelines and outcomes. The comparison factors can also be used to evaluate other authentication systems.

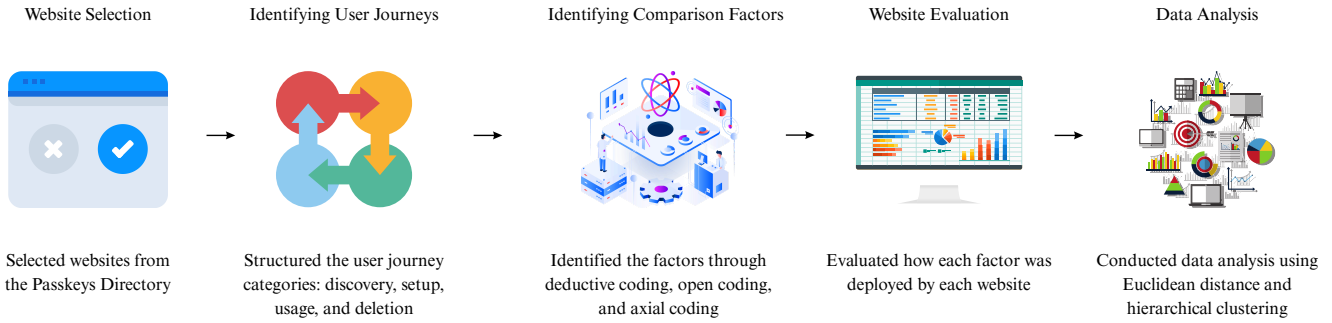


Figure 1: Methodology Overview

3 Methodology

Our methodology is shown in Figure 1. We selected 111 websites from `passkeys.directory` provided by 1Password (Section 3.1). We then identified and recorded passkey user journeys, categorizing them into discovery, setup, usage, and deletion stages (Section 3.2). We identified 28 comparison factors through deductive, open, and axial coding. We assessed how each website implemented these factors, classifying the implementations as either a match, partial match, or non-match (Section 3.3). We analyzed the data to measure the consistency of the passkey user experience and to identify similarities or differences in implementation patterns across the websites (Section 3.5).

We conducted all evaluations from February 2025 to February 2026 on a Windows 11 desktop running Google Chrome v134, connected to a university network. We used Android v13 for cross-device and cross-platform evaluations between the web and mobile app versions of a website. This standardized setup ensured consistent testing conditions and helped minimize the impact of external factors, such as risk-based authentication and regional differences.

To ensure reliability, two researchers independently evaluated and coded each website. They compared findings and resolved discrepancies through discussion to reach consensus on passkey user journeys, comparison factors, and website evaluations. This approach minimized personal bias, ensuring that codes were grounded in data rather than assumptions.

3.1 Website Selection

We selected the websites in our dataset from `passkeys.directory` [1], a community-driven index of websites, applications, and services supporting passkeys. We merged some domains after verifying that they shared the same authentication account. For example, `microsoft.com`, `aad.portal.azure.com`, and `entra.microsoft.com` all use the same authentication account, so we selected `microsoft.com` to represent these sites.

We excluded websites that required personal details, such as Social Security numbers, birthdates, and addresses, especially on financial and government websites. We also excluded sites from our dataset due to language barriers or registration fees. This process narrowed the selection to 111 websites from the 224 listed as of June 2025.

3.2 Identifying User Journeys

We mapped passkey user journeys by exploring and recording all potential ways users might navigate a website to set up, use, and delete passkeys on both mobile and desktop platforms. These journey explorations were informed by usability research on passkeys conducted by the FIDO Alliance User Experience Working Group, which identified 270 key moments across enterprise and consumer journeys that highlight significant user interactions with passkeys [41] as well as prior research on two-factor authentication by Lyastani et al. [32] and passkey usability research by Lassak et al. [29]. We categorized the passkey user journeys into four main stages: discovery, setup, usage, and deletion, while considering both new and existing users.

In the **discovery** stage, we analyzed how websites highlight their support for passkeys. We examined UI features such as links to documentation, banners, accordions, and displays that educate users about passkeys and facilitate setup. We also assessed how passkeys are presented relative to other authentication methods, focusing on whether they are prioritized and promoted as more secure options. Additionally, we reviewed the naming conventions used for passkey-based authentication to determine whether they adhere to FIDO Alliance recommendations.

In the **setup** stage, we analyzed the initial experience of configuring passkeys, whether users register via passwords, other authentication methods, or already possess a passkey. We compared the account registration process with passkeys to other authentication options. Additionally, we reviewed passkey setup across various website sections, including account settings, 2FA workflows with passwords and one-time passcodes, and account recovery. We also assessed the user experience of establishing multiple passkeys for a single ac-

count across different authenticators and how websites assist users in managing them.

Within the **usage** stage, we examined different processes for signing in to an account using passkeys. We also explored how websites support passkeys alongside other authentication methods and the priority assigned to passkeys in those scenarios. Additionally, we assessed the user experience of managing multiple passkeys for a single account across various authenticators, as well as how websites help users manage them. Furthermore, we compared the user experience of passkeys across web platforms and mobile versions.

Under the **deletion** stage, we analyzed how websites allow users to remove passkeys from their account settings. We checked if the option to delete a passkey was available within the passkey card affordance in account settings. Additionally, we assessed whether websites informed users to remove their passkeys from credential managers. We also evaluated the user experience of removing a passkey, focusing on confirmation dialogs, account lockout warnings, and user notifications.

3.3 Identifying Comparison Factors

We identified comparison factors through a two-stage coding process that combined deductive coding of existing passkey design guidance with inductive coding of observed user journeys. First, the primary author reviewed the FIDO Alliance Design Guidelines, including the required, optional, and security-key design patterns, and performed the initial deductive coding. For each design pattern, we identified the underlying guidelines and translated them into comparison factors that could be evaluated across websites.

For example, within the *Create, View, and Manage Passkeys in Account Settings* design pattern, we coded several guidelines as comparison factors. We coded the guideline *Allow people to create, view, and manage passkeys in Account Settings* as *Settings-Setup*. We coded the guideline *include the passkey hero prompt at the top of Account Settings* when users do not yet have a passkey, as *Promotion*. We coded the guideline *If passkeys exist for the user, allow them to see each passkey as a card* as *Management-UI*, the guideline *Place the passkeys UI above passwords in the interface* as *Priority*, and the guideline *Offer the ability to remove them* as *Delete-Option*.

When the interpretation or scope of a deductively derived factor was unclear, the main author clarified the coding process by discussing it with coauthors until they reached consensus. This deductive process produced 23 comparison factors based on the FIDO Alliance Design Guidelines. However, because real-world implementations may include recurring practices not fully captured by existing guidelines, we conducted a second coding step using the recorded user journeys. Two researchers independently applied open and axial coding to these journeys [30].

During open coding, the researchers examined the recorded user journeys and assigned conceptual codes to recurring implementation patterns, interaction points, and design choices. During axial coding, they grouped related codes to identify broader patterns across websites. This step captured recurring passkey implementation practices that were not fully represented by the guideline-derived factors. Disagreements were resolved through discussion, resulting in five additional comparison factors. Appendix A contains the codebooks showing how each comparison factor was derived.

3.4 Website Evaluation

After establishing the comparison factors, two researchers evaluated every website against them. We classified each implementation as fully matching, partially matching, not matching, or not applicable. A partial match indicated that a website satisfied some, but not all, of a factor's requirements when the factor could reasonably be evaluated in parts. Any disagreements were resolved through discussion to reach a consensus.

Our evaluation focused on the observable functionality of websites' passkey implementations rather than UI design, content quality, or usability. For example, we assessed whether a website provided an option to delete a passkey, but not whether the icons, layout, or other design choices made that option easy to find or use.

3.5 Data Analysis

We used Euclidean distance to quantify similarities in passkey implementations across various websites. It is calculated as the square root of the sum of squared differences between corresponding vector components [11]. In this context, the Euclidean distance between two websites indicates how much their implementation practices differ.

Each website was represented as a vector of ordinally encoded comparison factor scores. For example, two websites X and Y could be represented as `[3133311333132310333331331131]` and `[3133310333132333313333331230]`. The scores captured implementation levels for each factor (e.g., `0 = not applicable, 1 = non-match, 2 = partial match, 3 = full match`), with higher values indicating greater conformity. We used the vectors to calculate the Euclidean distance between website pairs X and Y.

To ensure precise comparisons, we used an adjusted Euclidean distance that considered only relevant, non-zero scores to prevent distortion from irrelevant data and capture both exact matches and weighted partial similarities.

We normalized the results to a 0-1 scale, organized them into a pairwise distance matrix, and applied agglomerative hierarchical clustering with the average linkage method [45] to identify groups of websites with similar passkey implementations. This clustering method merged clusters based on

average distances, revealing natural groupings and differences across sites.

For example, suppose three websites are evaluated across three factors, where 0 indicates that a factor is not applicable: $A = [0,2,3]$, $B = [1,2,2]$, and $C = [2,1,1]$. We compute distances pairwise, comparing one website vector with another at a time. When comparing A and B, the first factor is ignored because it is not applicable for A, so the adjusted Euclidean distance is computed only over the second and third factors:

$$d(A,B) = \sqrt{\frac{(2-2)^2 + (3-2)^2}{2}} = 0.71.$$

Similarly, the distance between A and C is:

$$d(A,C) = \sqrt{\frac{(2-1)^2 + (3-1)^2}{2}} = 1.58,$$

while the distance between B and C is:

$$d(B,C) = \sqrt{\frac{(1-2)^2 + (2-1)^2 + (2-1)^2}{3}} = 1.00.$$

We stored the pairwise distances in a distance matrix and normalized the matrix by dividing each value by the maximum observed distance, yielding 0–1 values for clustering and summary analysis. This approach prevents not-applicable factors from distorting similarity while preserving differences among non-match, partial-match, and full-match scores.

We selected Euclidean distance as a pragmatic similarity measure because our feature vectors include ordinal scores representing different levels of implementation: non-match, partial match, and full match. This allowed us to treat partial matches as intermediate between non-matches and full matches, rather than simply as "different," as Hamming distance would. We used equal weights as a simplifying assumption, since we did not have a theoretical model to justify unequal weights. Equal spacing and equal weighting are methodological assumptions and limitations of our analysis. However, prior work has found that converting ordinal data to numeric values can sometimes perform better than more complex linear models [9].

3.6 Limitations

This study has certain limitations affecting how broadly its findings can be applied. It did not cover the full spectrum of user journeys, such as accessibility-focused scenarios, non-traditional devices like smart TVs, gaming consoles, or in-car systems, or native mobile apps. Passkey behavior may vary significantly between desktop browsers, mobile browsers, and native applications. Furthermore, our dataset may be biased, as it partly relies on global website rankings and `passkeys.directory`, which likely overrepresent large, English-language services with advanced security practices. Therefore, we cannot confidently generalize our findings to all websites. Additionally, adaptive or risk-based authentication mechanisms may follow different processes depending on the context, something our controlled evaluation could not fully capture. Passkey implementations also change quickly, so our results represent a specific point in time.

Our evaluation was conducted on Windows 11 with Google Chrome, which may limit the generalizability of our results. For instance, early in testing, a researcher using Linux encountered a website that refused to show its login interface on their device. We selected Windows 11 and Chrome because they are widely used and well supported [47, 48], providing a consistent environment for website testing. Future research should explore passkey experiences on less-supported yet popular platforms—such as Linux, various browsers, mobile browsers, and native mobile apps.

While our coding scheme includes partial matches to capture implementation nuances beyond simple match or no-match decisions, it still simplifies the variability seen in real-world passkey implementations. A partial match could indicate varying degrees or types of incompleteness across websites; for instance, one site may meet most of the elements of a guideline, while another only satisfies a few, yet both get the same partial-match score. Additionally, two sites might be coded as partial matches for different reasons, causing them to seem more similar in distance analysis than they are in reality. Since our analysis treats partial matches as a single ordinal category, it may hide subtle differences in implementation quality, usability, and security implications. Future research might improve this by creating factor-specific rubrics, weighted scores, or qualitative notes to better differentiate between diverse forms and levels of partial implementation.

Finally, while Euclidean distance is a useful way to measure differences in passkey implementation patterns across websites, it has limitations. First, it assumes that the encoded scores have meaningful numeric spacing, which may oversimplify categorical judgments such as non-match, partial match, and full match. Second, although the metric reflects overall similarity, it does not clarify why websites differ or identify which specific design choices are most responsible for those differences. Therefore, Euclidean distance should be viewed as a high-level measure of similarity and should be complemented by factor-level analysis and user-centered studies to better understand the practical implications of the observed differences.

3.7 Ethics Statement

We are committed to fairness, transparency, and responsible research practices. Although our study was not reviewed by our IRB as we did not perform research involving human subjects or collect identifiable personal data, we adhere to the following ethical research practices. To respect website operators and users, we obtained all data from publicly ac-

cessible sources and did not collect proprietary or sensitive information. Following beneficence and justice, we generated minimal network traffic during testing and chose to evaluate websites using the most widely supported operating system and browser combination to provide websites with a consistent and widely supported testing environment. To reduce potential reputational harm, we also do not present any websites as negative examples, though we do present our full dataset for transparency.

4 Results

Next, we present the results of our study, including the comparison factors that form the basis of our comparative framework, as outlined in Section 4.1. We also present the results of the website evaluation against these factors in Section 4.2. Lastly, we present the results of our analysis of the consistency of the passkey user experience, using Euclidean distance and hierarchical clustering in Section 4.3.

4.1 Comparison Factors

The following factors include those we identified through deductive coding and those we derived through subsequent open and axial coding of passkey implementations. We annotate each deductively-coded factor with its corresponding type of FIDO Alliance Design Guidelines pattern: required patterns (RP), optional patterns (OP), and security key patterns (SP). We designated the factors derived through open and axial coding as additional patterns (AP). When evaluating a website, we considered a factor a match if it was implemented as described; otherwise, we classified it as a partial match, non-match, or not-applicable when those categories applied. Together, these factors describe recommended or commonly observed practices for implementing passkey features.

4.1.1 Factors for Discovery

These factors cover passkey promotion through documentation, help links, and educational materials.

F01. Promotion (RP): The website clearly shows its passkey support during setup or in account settings, using elements like a banner, pop-up, or a message with a title, a brief description, and a call to action such as an "Add Passkeys" button.

F02. Priority (RP): The website places the passkey user interface above passwords and other authentication options. For instance, in the account settings, the passkey interface is displayed prominently above the password section, where both options are available.

F03. HSK-Awareness (SK): The website informs users that passkeys on hardware security keys (HSKs), such as YubiKeys, are valid authentication methods. It provides this information through different communication channels and resources, including documentation, a dedicated passkey feature, pop-up alerts, and prominently displayed messages.

F04. Educational-Resources (AP): The website offers detailed passkey information through links, such as a "learn more" link, or interface elements, such as dropdown menus. We consider this a partial match if users must search the website's knowledge base or support areas to find this information.

F05. Naming-Conventions (AP): The passkeys interface of the website is labeled with the term "Passkey(s)" and accompanied by approved FIDO logo trademarks or passkey icons.

4.1.2 Factors for Setup

These factors cover passkey setup.

F06. Registration-Setup (OP): The website encourages users to set up a passkey when they create an account. If the website asks the user to add a passkey immediately after registration, we consider this a partial match.

F07. OTP-Setup (OP): The website shows passkeys to users immediately after password and SMS One-Time Passcode (OTP) authentication, if available. It also allows users to create a passkey instead of using passwords and SMS OTPs, or presents passkeys as a more secure way to authenticate. We consider it not applicable if the website does not use passwords or support 2FA via SMS OTP.

F08. Settings-Setup (RP): The website allows users to add passkeys via their account settings.

F09. HSK-Setup (SP): The website allows users to add passkeys via hardware security keys like YubiKeys.

F10. Cross-Device-Setup (OP): The website allows users to create a passkey on various mobile devices, including smartphones, tablets, and iPads. As a result, users can switch between devices without needing to reconfigure passkeys.

F11. Recovery-Setup (OP): When users forget their password, the website lets them add a passkey as a replacement or addition during account recovery. After verifying their identity, users can select to create either a passkey or a new password. If they choose a new password, the website also supports passkey creation. If the website does not use passwords, this factor is not applicable.

F12. Multicreation (OP): The website lets users create multiple passkeys of various types, using either platform or roaming authenticators.

F13. Setup-Authentication (AP): Users are required to verify their identity on the website before adding a passkey to their account. However, we consider it a partial match if the website only asks for user verification once when accessing account settings.

F14. Setup-Notification (OP): The website notifies users of the success or failure of their passkey addition via email, push notifications, or SMS.

F15. Update-Option (AP): The website lets users customize their passkeys, including renaming them according to

their preferences. If there is no passkey management interface, we consider this factor not applicable.

F16. Update-Notification (OP): The website notifies users of passkey update success or failure via email, push notifications, or SMS. If the website does not offer an update option, we consider this factor not applicable.

4.1.3 Factors for Usage

These factors cover the practical use of passkeys on websites.

F17. Sign-In-UI (RP): The website enables users to sign in with a passkey by clicking a designated button or link to initiate authentication. It either features a "Sign in with passkeys" interface or lacks an autofill feature but still allows signing in through the "Next" or "Sign In" buttons.

F18. Autofill-Feature (RP): The website allows users to sign in with a passkey through the autofill feature, which suggests a passkey, if available, in the identifier field of the sign-in interface.

F19. Fallback-Authentication (RP): The website provides extra authentication options beyond passkeys, including passwords and social logins like Apple, Facebook, or Google.

F20. Cross-Device-Usage (OP): The website login UI accepts passkeys from various mobile devices, including smartphones, tablets, or iPads. We consider it a partial match if the website does not support cross-device setup (F10).

F21. HSK-Usage (SP): The website login UI accepts passkeys on hardware security keys. We consider it a partial match if the website does not support HSK-Setup (F9).

F22. Mobile-App-Usage (OP): The website allows users to access their accounts via both a website and a mobile app using the same passkey. If a mobile app is not offered, we consider this factor not applicable.

F23. Management-UI (RP): The website provides a passkey management interface under a section title "Passkeys", where users can view and manage registered passkeys.

4.1.4 Factors for Deletion

These factors cover removing passkeys in settings and notifying users of those changes.

F24. Delete-Option (RP): The website allows users to delete their passkeys.

F25. Authenticator-Cleanup (OP): The website recommends users delete the removed passkey from their authenticator. If the site does not provide an option to delete a passkey, we consider this factor not applicable.

F26. Deletion-Authentication (AP): The website asks users to verify their identity in order to remove a passkey from their account. If verification is only needed to access security settings, we consider this a partial match. If there is no option to delete a passkey, we consider this factor not applicable.

F27. Deletion-Notification (OP): The website informs users about the success or failure of passkey deletion via multiple channels, including email, push notifications, and SMS. If the option to delete a passkey is not available on the website, this factor is considered not applicable.

F28. Deletion-Confirmation (OP): The website confirms the user's intent with a confirmation dialog before deleting a passkey. When the last passkey is removed, the website either informs the user that they will need to set up another form of authentication for their account or prevents the removal of the last passkey, as appropriate. If either of these parameters is missing, we consider it a partial match. If the option to delete a passkey is not available on the website, this factor is considered not applicable.

4.2 Website Factors Evaluation

Table 1 presents our website evaluation. We group the websites into three clusters based on similar passkey implementation patterns, ordered by website rank within each cluster. Table 2 presents the comparison factors grouped by pattern type, ordered by implementation rate within each type. Table 3 presents the implementation rates for each factor within every cluster. The percentages in Table 2 and Table 3 include both full and partial matches, and exclude not-applicable cases.

4.3 Consistency Measurements

We used Euclidean distance and hierarchical clustering with the average linkage method to evaluate the consistency of the passkey user experience. Furthermore, we analyzed each website cluster to identify similarities in how passkeys are implemented.

4.3.1 Euclidean Distance

We evaluated the similarity of passkey implementation practices across different websites by calculating pairwise Euclidean distances within our dataset. We then summarized these distances using the mean, median, and mode to represent typical and central levels of similarity. On a scale from 0 (most similar) to 1 (least similar), the mode, mean, and median distances were 0.55, 0.57, and 0.56, respectively.

As illustrated in Figure 2, the mode is located near both the mean and median, indicating a moderate degree of similarity in passkey implementation practices across websites. Figure 2 presents a histogram depicting the distribution of pairwise Euclidean distances among all website pairs. This histogram shows the frequency of various similarity levels and provides insight into whether the dataset predominantly consists of highly similar, moderately similar, or diverse implementations.

Table 1: Website Evaluation Results

Ranked Websites	Discovery					Setup										Usage							Deletion				
	F01: Promotion	F02: Priority	F03: HSK-Awareness	F04: Educational-Resources	F05: Naming-Conventions	F06: Registration-Setup	F07: OTP-Setup	F08: Settings-Setup	F09: HSK-Setup	F10: Cross-Device-Setup	F11: Recovery-Setup	F12: Multicreation	F13: Setup-Authentication	F14: Setup-Notification	F15: Update-Option	F16: Update-Notification	F17: Sign-In-UI	F18: Autofill-Feature	F19: Fallback-Authentication	F20: Cross-Device-Usage	F21: HSK-Usage	F22: Mobile-App-Usage	F23: Management-UI	F24: Delete-Option	F25: Authenticator-Cleanup	F26: Deletion-Authentication	F27: Deletion-Notification	F28: Deletion-Confirmation
Cluster 1: Standard Implementers (n = 89)																												
1: Google	●	●	●	●	●			●	●	●		●		●	●		●	●	●	●	●	●	●	●			●	●
3: Microsoft	●		●	●	●			●	●	●		●		●	●	●	●		●	●	●		●	●			●	◐
18: LinkedIn	●			●	●			●	●	●		●	●			■		●	●	●	●	●	●	●	●	●		◐
26: Amazon	●	●	●	●	●	◐		●	●	●		●		●		■	●		●	●	●	●	●	●	●		●	◐
35: Github	●		●	●	●			●	●	●		●		●	●		●	●	●	●	●	●	●	●	●		●	◐
52: Yahoo					●	●		●	●	●		●		●		■	●		●	●	●		●	●			●	
62: Adobe	●		●	●	●			●	●	●		●	◐	●		■	●	●	●	●	●		●	●			●	●
65: Roblox	●	●	●	●	●		■	●	●	●		●	●	●		■		●	●	●	●	●	●	●	●	●	●	◐
113: VK	●		●	●	●			●	●	●		●		●	●	●	●		●	●	●	●	●	●		●		◐
188: Ebay	●	●		◐	●	◐		●	●	●		●		●		■	●		●	●	●		●	●			●	◐
216: Discord	●							●	●	●		●	●	●	●		●	●	●	●	●	●	●	●		●		■
235: Shopify	●	●		●	●		■	●	●	●		●	●	●		■	●		●	●	●	●	●	●		●	●	◐
240: Canva	●				●		■	●	●	●	■	●	●	●		■	●		●	●	●	●	●	●		●	●	◐
253: Hubspot	●			●	●			●	●	●		●		●	●		●		●	●	●	●	●	●		●	●	◐
284: Stripe	●		●	●	●		■	●	●	●		●	●	●	●		●		●	●	●		●	●	●	●	●	◐
406: Nvidia			●				■	●	●	●		●		●	●	●	●		●	●	●	●	●	●			●	◐
451: Uber	●				●		●	●	●	●	■	●		●		■	●	●	●	●	●	●	●	●	●		●	◐
566: PlayStation	●	●		●	●			●	●	●	●	●		●		■		●	●	●	●	●	●	●			●	●
834: Zoho	●			●	●			●	●	●		●		●		■	●		●	●	●	●	●				●	◐
856: Synology	●						■	●	●	●		●	●	●	●	●	●	●	●	●	●	●	●	●		●		◐
1167: Notion	●				●			●	●	●		●		●	●		●		●	●	●	●	●	●			●	◐
1770: Coinbase	●		●	●	●			●	●	●		●	●	●	●	●	●		●	●	●		●	●		●	●	◐
1959: hCaptcha	●	●	●		●		■	●	●	●		●		●	●	●		●	●	●	●	■	●	●	●		●	◐
1969: Hyatt	●	●	●	●	●		■	●	●	●	■	●		●		■		●	●	●	●			●			●	◐
2149: Square Up	●			●	●			●	●	●		●		●	●	●	●		●	●	●		●	●	●		●	◐
2475: Hatena	●			●	●	◐	■	●	●	●		●		●	●	●	●	●	●	●	●	●	●	●			●	◐
2495: Home Assistant	●				●		■	●	●	●		●		●	●		●	●	●	●	●		●	●				◐
3319: Porkbun	●		●	●	●	●	■	●	●	●	●	●				■	●		●	●	●	■	●	●				◐
3815: Bitwarden	●		●	●	●		■	●	●	●	■	●	●	●		■	●		●	●	●		●	●		●	●	◐
4345: Docusign	●		●	●	●		■	●	●	●			◐	●	●	●	●		●	●	●	●	●	●		◐	●	●
4914: Kayak	●				●	●	■	●	●	●	■	●		●		■		●	●	●	●	●	●	●			●	◐
5091: Arbeitsagentur	●			●	●		■	●	●	●		●		●	●	●	●	●	●	●	●	●	●	●	●		●	◐
6158: Visma	●	●	●		●		■	●	●	●		●	●	●		■	●		●	●	●	●	●	●		●	●	◐
8212: Trimble	●				●	●		●	●	●		●	●	●	●		●		●	●	●	●	●	●		●	●	◐
11785: Fastmail	●				●			●	●	●		●		●		■		●	●	●	●	●	●	●				
12585: Discourse	●				●		■	●	●	●		●	●	●	●		●	●	●	●	●		●	●		●		◐
12687: Vercel	●	●	●	●	●		■	●	●	●	■	●		●	●	●	●		●	●	●	■	●	●			●	◐
12848: Larksuite	●		●	●	●			●	●	●		●	●	●	●	●	●		●	●	●	●	●	●		●	●	◐
13139: BMW	●			●	●		■	●	●	●		●	●	●	●	●	●		●	●	●	●	●	●			●	◐
13450: Microcenter	●				●		■	●	●	●		●			●		●		●	●	●	■	●	●				◐
15888: Ankiweb	●				●		■	●	●	●		●		●			●	●	●	●	●		●	●				◐
17014: Esl Gaming	●		●		●		■	●	●	●		●				■		●	●	●	●	■	●	●				●
19105: Linear	●	●			●		■	●	●	●	■	●		●	●		●		●	●	●	●	●	●			●	◐
19377: Migros	●			●	●		■	●	●	●		●		●		■	●	●	●	●	●	■	●	●	●		●	◐
24116: Crowdin	●		●		●		■	●	●	●		●		●		■	●		●	●	◐	■	●	●			●	◐
29064: XenForo	●	●			●		■	●	●	●		●		●	●		●		●	●	●	■	●	●				◐
30582: Spaceship	●	●	●	●	●	●	■	●	●	●		●		●	●	●	●	●	●	●	●	■	●	●	●		●	●
34119: FusionAuth	●	●			●		■	●	●	●		●				■	●		●	●	●	■	●	●				◐
36976: Luno	●				●		■	●	●	●		●		●		■		●	●	●	●	●	●	●			●	●
51716: Devolutions	●		●		●			●	●	●		●	●	●	●		●		●	●	●		●	●		●		
52084: Moneybird	●		●	●	●	◐	■	●	●	●		●	●	●	●		●	●	●	●	●	●	●	●	●	●	●	◐
60863: Level1Techs	●				●		■	●	●	●		●	◐	●	●		●	●	●	●	●	■	●	●		◐		◐
71872: Air New Zealand	●	●	●	●	●		■	●	●	●	■	●	●	●	●			●	●	●	●	●	●	●	●	●	●	◐
83099: Nulab	●		●	●	●		■	●	●	●		●		●	●	●	●	●	●	●	●	■	●	●	●		●	●
93748: Dinero	●	●	●		●		■	●	●	●		●	●	●		■	●		●	●	●	●	●	●		●	●	◐
99529: Cloze					●		■	●	●	●		●	●	●		■	●	●	●	●	●	●	●	●		●	●	◐
167117: Finom	●	●	●	●	●		■	●	●	●		●	●	●	●	●	●		●	●	●		●	●		●	●	

Legend: ● = *factor matches* ◐ = *factor partially-matches* □ = *factor does not match* ■ = *factor not applicable*

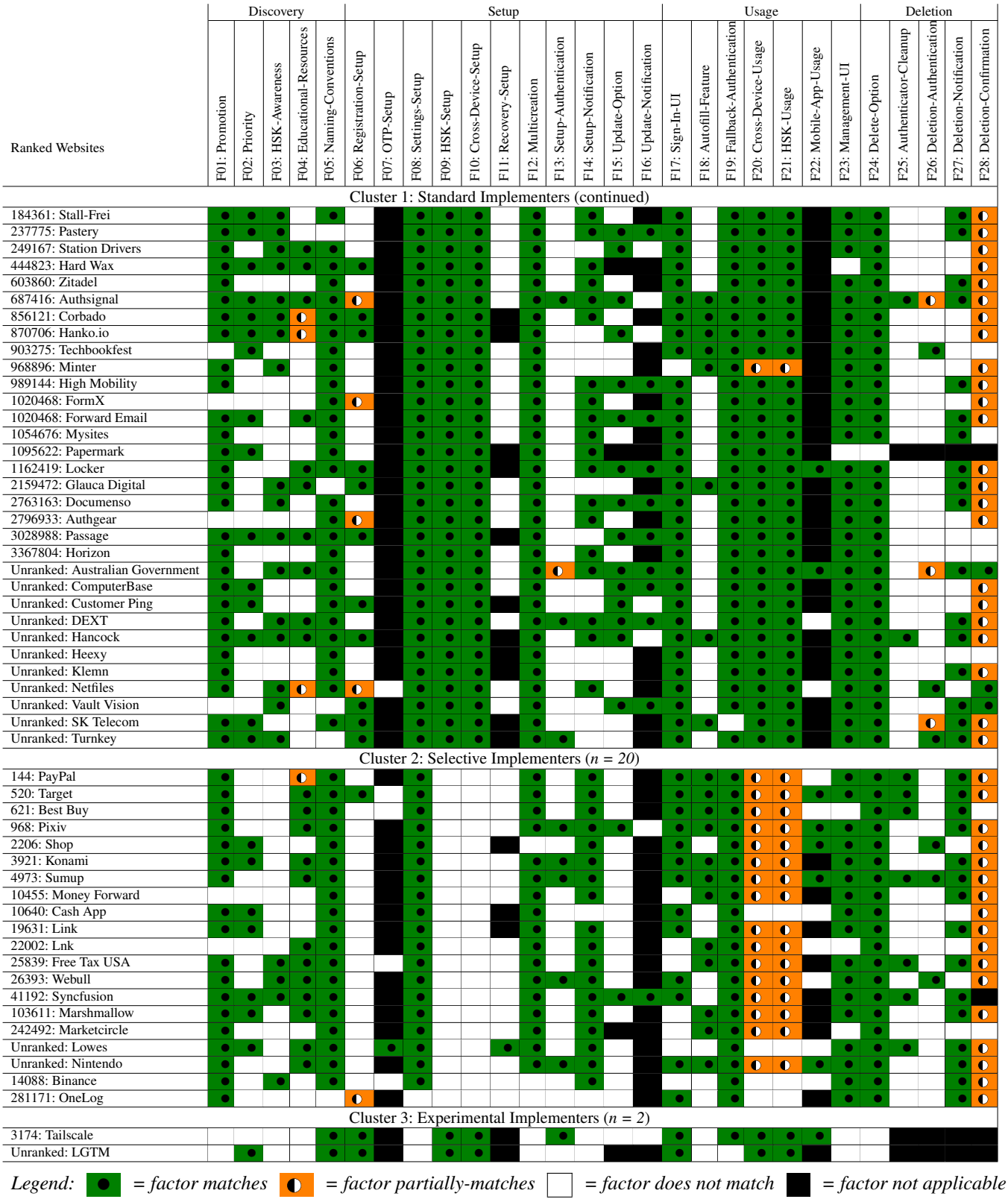

Discovery
Setup
Usage
Deletion
Ranked Websites
F01: Promotion
F02: Priority
F03: HSK-Awareness
F04: Educational-Resources
F05: Naming-Conventions
F06: Registration-Setup
F07: OTP-Setup
F08: Settings-Setup
F09: HSK-Setup
F10: Cross-Device-Setup
F11: Recovery-Setup
F12: Multicreation
F13: Setup-Authentication
F14: Setup-Notification
F15: Update-Option
F16: Update-Notification
F17: Sign-In-UI
F18: Autofill-Feature
F19: Fallback-Authentication
F20: Cross-Device-Usage
F21: HSK-Usage
F22: Mobile-App-Usage
F23: Management-UI
F24: Delete-Option
F25: Authenticator-Cleanup
F26: Deletion-Authentication
F27: Deletion-Notification
F28: Deletion-Confirmation
Cluster 1: Standard Implementers (continued)
184361: Stall-Frei
237775: Pastery
249167: Station Drivers
444823: Hard Wax
603860: Zitadel
687416: Authsignal
856121: Corbado
870706: Hanko.io
903275: Techbookfest
968896: Minter
989144: High Mobility
1020468: FormX
1020468: Forward Email
1054676: Mysites
1095622: Papermark
1162419: Locker
2159472: Glauca Digital
2763163: Documenso
2796933: Authgear
3028988: Passage
3367804: Horizon
Unranked: Australian Government
Unranked: ComputerBase
Unranked: Customer Ping
Unranked: DEXT
Unranked: Hancock
Unranked: Heexy
Unranked: Klemn
Unranked: Netfiles
Unranked: Vault Vision
Unranked: SK Telecom
Unranked: Turnkey
Cluster 2: Selective Implementers (n = 20)
144: PayPal
520: Target
621: Best Buy
968: Pixiv
2206: Shop
3921: Konami
4973: Sumup
10455: Money Forward
10640: Cash App
19631: Link
22002: Lnk
25839: Free Tax USA
26393: Webull
41192: Syncfusion
103611: Marshmallow
242492: Marketcircle
Unranked: Lowes
Unranked: Nintendo
14088: Binance
281171: OneLog
Cluster 3: Experimental Implementers (n = 2)
3174: Tailscale
Unranked: LGTM
Legend: = factor matches = factor partially-matches = factor does not match = factor not applicable

Table 2: Implementation rates of the comparison factors

Comparison Factors	Stage	Rates
Required Patterns		
F19: Fallback-Authentication	Usage	98%
F08: Settings-Setup	Setup	97%
F24: Delete-Option	Deletion	97%
F23: Management-UI	Usage	93%
F01: Promotion	Discovery	90%
F17: Sign-In-UI	Usage	84%
F18: Autofill-Feature	Usage	42%
F02: Priority	Discovery	36%
Optional Patterns		
F20: Cross-Device-Usage	Usage	96%
F12: Multicreation	Setup	95%
F28: Deletion-Confirmation	Deletion	91%
F10: Cross-Device-Setup	Setup	82%
F14: Setup-Notification	Setup	80%
F27: Deletion-Notification	Deletion	69%
F22: Mobile-App-Usage	Usage	63%
F16: Update-Notification	Setup	53%
F06: Registration-Setup	Setup	25%
F25: Authenticator-Cleanup	Deletion	23%
F07: OTP-Setup	Setup	7%
F11: Recovery-Setup	Setup	3%
Security Key Patterns		
F21: HSK-Usage	Usage	96%
F09: HSK-Setup	Setup	82%
F03: HSK-Awareness	Discovery	43%
Additional Patterns		
F05: Naming-Conventions	Discovery	93%
F04: Educational-Resources	Discovery	51%
F15: Update-Option	Setup	47%
F13: Setup-Authentication	Setup	30%
F26: Deletion-Authentication	Deletion	30%

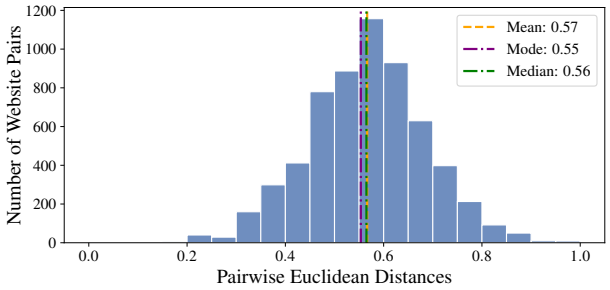


Figure 2: Distribution of Euclidean Distances

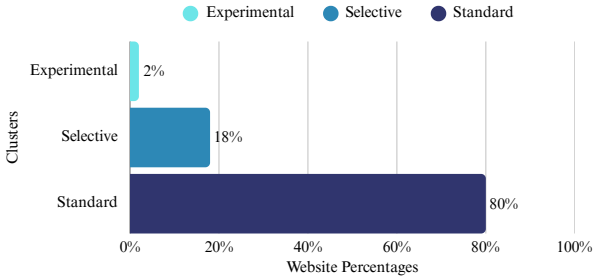


Figure 3: Clusters of Websites

4.3.2 Hierarchical Clustering

We used agglomerative hierarchical clustering with the average-linkage method to group websites by their passkey implementation practices. Figure 3 shows the results of this clustering, dividing the websites into three clusters: standard, selective, and experimental implementers. Overall, our results suggest that most websites follow similar passkey implementation practices, with a few exceptions that have some unique traits, discussed below.

Most websites (80%) are in Cluster 1, which we call Standard Implementers because they excel at promoting passkeys (F01), supporting multiple passkeys across different devices (F08, F20, F21, F12), and managing passkeys effectively (F23, F24) as shown in Table 3. Cluster 2, which includes 18% of the websites, is labeled Selective Implementers because these sites have practices similar to those of the Standard Implementers but do not support cross-device passkey setup (F10) or passkey setup on hardware security keys (F09). Cluster 3 contains just two sites; we label these Experimental Implementers, as they have weaker passkey management. These sites provide incomplete support for passkey setup, use, and deletion, and neither promote passkeys nor offer educational content. Interestingly, they support creating new accounts with passkeys across a variety of devices (F06, F10, and F20) and on FIDO security keys (F09, F21), though only one passkey per account is allowed.

Table 3 shows that the Standard Implementers cluster has the highest average implementation rates (67%) compared to the other groups. The Selective Implementers, which include sites with partial or phased implementation, have an average implementation rate of around 54%. While some features are well-developed, these sites appear to be in an intermediate stage of passkey adoption, focusing on core functionalities rather than on all optional and security features outlined in the FIDO Alliance Design Guidelines. On the other hand, the Experimental Implementers consist of sites with limited or experimental passkey integrations, with most factors having low implementation rates and an overall average of 34%. This may indicate that they are still in initial testing phases or exploring how to implement passkeys. Overall, we observed that most websites consistently implement most comparison factors and adapt each factor to their specific business needs.

Table 3: Implementation Rates per Cluster

Factor	Standard	Selective	Experimental
F01: Promotion	93%	90%	0%
F02: Priority	36%	35%	50%
F03: HSK-Awareness	50%	20%	0%
F04: Educational-Resources	50%	65%	0%
F05: Naming-Conventions	93%	95%	100%
F06: Registration-Setup	27%	10%	100%
F07: OTP-Setup	5%	20%	0%
F08: Settings-Setup	100%	95%	0%
F09: HSK-Setup	100%	0%	100%
F10: Cross-Device-Setup	100%	0%	100%
F11: Recovery-Setup	3%	6%	0%
F12: Multicreation	99%	85%	0%
F13: Setup-Authentication	31%	25%	50%
F14: Setup-Notification	80%	90%	0%
F15: Update-Option	56%	11%	0%
F16: Update-Notification	55%	50%	0%
F17: Sign-In-UI	88%	65%	100%
F18: Autofill-Feature	40%	60%	0%
F19: Fallback-Authentication	99%	100%	50%
F20: Cross-Device-Usage	100%	80%	100%
F21: HSK-Usage	100%	80%	100%
F22: Mobile-App-Usage	68%	42%	100%
F23: Management-UI	97%	85%	0%
F24: Delete-Option	99%	100%	0%
F25: Authenticator-Cleanup	21%	35%	0%
F26: Deletion-Authentication	33%	15%	0%
F27: Deletion-Notification	68%	75%	0%
F28: Deletion-Confirmation	91%	90%	0%
Average	**67%**	**54%**	**34%**

5 Discussion

5.1 Passkey User Experience Consistency

Our findings show that core passkey support is converging, but supporting features remain uneven. As shown in Table 1, 80% of websites fall into Cluster 1 (Standard Implementers), indicating broadly similar implementation patterns across most websites in our dataset.

A distinguishing characteristic of Cluster 2, the Selective Implementers, is a mismatch between passkey enrollment and passkey use across authenticator types. None of the websites in this cluster support creating passkeys with hardware security keys or mobile devices, as reflected by the absence of matches for HSK-Setup (F09) and Cross-Device-Setup (F10). However, most of these websites still allow users to sign in with those authenticators, as shown by the partial matches for Cross-Device-Usage (F20) and HSK-Usage (F21).

This mismatch suggests that some websites rely on default options provided by passkey libraries during authentication, which may expose available authenticator transport options through `allowCredentials` entries in `PublicKeyCredentialRequestOptions` [36] without providing corresponding enrollment flows for those same authenticators. These websites might confuse users by appearing to support various devices without a way to register them.

Cluster 3 is characterized by a large number of non-matching factors, particularly the absence of passkey setup in account settings and passkey deletion support (F08, F24), leaving the remaining deletion factors not applicable (F25–F28). Beyond these cluster-level patterns, several lower-adoption factors highlight areas where passkey implementations remain uneven. We noticed some inconsistencies between the web and mobile app platforms. For 21% of websites in our dataset, passkeys created on the website could not be used to authenticate in the corresponding mobile app, as shown by websites coded as non-matches for F22. This could cause confusion about whether the passkey was created correctly and may also diminish the perceived portability and convenience of passkeys.

Other unevenly implemented factors affect how easily users discover and use passkeys. Passkeys are not consistently prioritized in account settings or authentication interfaces (F02), with only 36% of websites placing passkey options above passwords or other authentication methods. The autofill feature (F18), which allows browsers and credential managers to surface available passkeys directly in the sign-in interface, was implemented by only 42% of websites.

Overall, these findings suggest that the passkey ecosystem is converging around basic availability: most websites in our dataset provide passkey setup, sign-in, management, and deletion through broadly similar patterns. However, convergence is weaker for the supporting features that make passkeys understandable, portable, and secure across the full account lifecycle. Inconsistencies in authenticator enrollment, cross-platform use, autofill support, passkey prioritization, recovery, and lifecycle authentication may limit users' ability to form stable expectations about when passkeys are available and how to use them. Addressing these gaps would help move passkey deployments from basic support toward a more reliable and trustworthy passwordless authentication experience.

5.2 Comparative Framework

Our comparative framework is the main contribution of this work. While our findings characterize the current state of passkey implementations, the framework provides a reusable methodology to study how passkeys are presented, configured, managed, and maintained across websites. The framework supports the systematic comparison of implementations, longitudinal measurement of ecosystem change, and factor-level analysis to guide future studies.

In this study, the framework enabled a systematic comparison of 111 websites based on 28 factors to measure the consistency of passkey user journeys. This comparison can inform future revisions to the FIDO Alliance Design Guidelines by identifying which patterns are widely adopted, inconsistently implemented, or frequently missing. For instance, low adoption of identity verification and recovery-related practices suggests a need for more concrete guidance on passkey

lifecycle security, while limited support for hardware security key awareness and authenticator cleanup points to gaps in guidance around device management. The framework also captures practices that appear in some implementations but are not currently emphasized in the guidelines, providing a structured basis for refining passkey guidance as the ecosystem evolves.

Future replication studies can build on this framework to measure how the passkey experience changes over time as websites update their authentication flows and the FIDO Alliance Design Guidelines evolve. Repeated evaluations could show whether the passkey ecosystem is becoming more standardized, more fragmented, or more unevenly adopted across industry sectors and website rankings. The framework could also support manual, semi-automated, or automated evaluation tools that check for the presence, absence, or partial support of passkey features.

The framework can also help researchers design targeted, in-depth studies of specific parts of the passkey user journey. Rather than treating passkey usability as one broad problem, researchers can use the framework to select factors that are underused, inconsistently implemented, or important for user understanding and account security. In this way, the framework provides a foundation for research on the usability, security, and privacy of passwordless authentication.

5.3 FIDO Alliance Design Guidelines

To enhance users' understanding of passkeys, the FIDO Alliance should develop standardized educational resources and interactive tools that websites can easily integrate. We observed that several websites lack clear, user-facing content explaining passkey benefits, workflows, and security features. Prior work highlights the importance of user education in increasing passkey adoption [27, 29, 32]; thus, centralized FIDO-approved educational modules, such as interactive demos, guides, and visual materials, could promote consistent messaging, improve user understanding, and lower adoption barriers. More generally, our results suggest that future revisions of the FIDO Alliance Design Guidelines could prioritize areas where websites remain inconsistent, particularly user education, authenticator guidance, and lifecycle transparency.

5.4 Future Work

Building on the framework-level opportunities described above, our results motivate three specific directions for future work: communication and comprehension, lifecycle security and usability studies, and replication across broader settings.

Although 90% of websites provided promotional materials (F01), we observed significant gaps in how they explained topics such as device loss, account recovery, fallback authentication, and the security and privacy tradeoffs of different authenticator types. Future research could examine how contextual explanations, educational resources, and guidance on hardware security keys (F03, F04) affect users' understanding of passkeys and their ability to make informed decisions.

Our results show that nearly 70% of websites do not reauthenticate users when adding or removing passkeys (F13, F26). This creates a potential lifecycle risk: an attacker with temporary access to an active session could register their own passkey to gain persistent access or delete existing passkeys to lock the user out of their account. Recent research on interpersonal threats has explored related risks with passkeys [8]. Future work could explore how to better protect lifecycle actions while balancing security with usability. Researchers could also examine whether notifications (F14, F16, F27), account activity logs, and similar mechanisms can help users review authenticator changes, track passkey use, detect unauthorized changes, and recover from attacks.

Future work could also test whether the patterns observed in this study hold across other platforms, devices, and browsers. In particular, longitudinal replication studies could assess whether the 28-factor framework captures convergence or fragmentation over time, while cross-platform studies could examine whether the same websites provide consistent experiences across desktop and native mobile applications.

6 Conclusion

Passkeys represent an important step toward more usable and phishing-resistant authentication, but their success depends not only on cryptographic strength but also on consistent, understandable, and secure implementation across the web. In this study, we evaluated passkey deployments across 111 websites using a comparative framework of 28 comparison factors spanning discovery, setup, usage, and deletion. Our framework supports longitudinal comparisons, automated evaluation tools, and studies targeting specific factors of interest.

Our findings show that the passkey user experience is moderately consistent, with most websites converging on core implementation patterns such as passkey setup, management, and deletion. However, this convergence remains uneven. Several supporting features, such as passkey education, cross-platform support, and identity verification for lifecycle actions, remain inconsistently implemented or underused. These gaps may limit the usability, security, and privacy benefits that passkeys are intended to provide by creating user confusion, increasing lifecycle security risks, and reducing confidence in passkeys across devices and platforms.

By introducing a factor-based framework for passkey deployments, this work provides a foundation for future replication studies, usability research, implementation audits, and revisions to passkey design guidelines. As passkeys continue to evolve, improving the passkey experience will be essential to ensure that they are not only technically secure but also reliable, understandable, and trustworthy for everyday users.

Acknowledgments

We thank Daniel Zappala and Xinru Page for helpful feedback and guidance during the early stages of this project.

Availability

Supplementary materials, including code used for consistency measurements and clustering, are available at https://zenodo.org/records/20709272.

References

[1] Passkeys.directory: A community-driven index of passkey-supporting services. https://passkeys.directory/, 2025. Accessed: 2026-01-12. URL: https://passkeys.directory/.

[2] Youssef Amer, Ehsan Ul Haque, Zhelun Rong, and Mohammad Maifi Hasan Khan. Understandability of the technology and benefit may not be enough to nudge users: An exploratory study in the context of FIDO2 adoption behavior. In *2025 IEEE 49th Annual Computers, Software, and Applications Conference (COMPSAC)*, pages 607–618, 2025. doi:10.1109/COMPSAC65507.2025.00083.

[3] Apple Inc. Passkeys. Overview of passkeys, their security benefits, and developer guidance on implementing passkey support in apps and websites. URL: https://developer.apple.com/passkeys/.

[4] Robert Biddle, Sonia Chiasson, and P.C. Van Oorschot. Graphical passwords: Learning from the first twelve years. *ACM Computing Surveys (CSUR)*, 44(4), September 2012. doi:10.1145/2333112.2333114.

[5] Joseph Bonneau, Cormac Herley, Paul C. van Oorschot, and Frank Stajano. The quest to replace passwords: A framework for comparative evaluation of web authentication schemes. In *IEEE Symposium on Security and Privacy*, pages 553–567, 2012. doi:10.1109/SP.2012.44.

[6] Joseph Bonneau and Sören Preibusch. The password thicket: Technical and market failures in human authentication on the web. In *The Workshop on the Economics of Information Security (WEIS)*, 2010.

[7] Stéphane Ciolino, Simon Parkin, and Paul Dunphy. Of two minds about Two-Factor: Understanding everyday FIDO U2F usability through device comparison and experience sampling. In *Fifteenth Symposium on Usable Privacy and Security (SOUPS)*, pages 339–356, August 2019. URL: https://www.usenix.org/conference/soups2019/presentation/ciolino.

[8] Alaa Daffalla, Arkaprabha Bhattacharya, Jacob Wilder, Rahul Chatterjee, Nicola Dell, Rosanna Bellini, and Thomas Ristenpart. A framework for abusability analysis: The case of passkeys in interpersonal threat models. In *34th USENIX Security Symposium*, pages 7819–7838, August 2025. URL: https://www.usenix.org/conference/usenixsecurity25/presentation/daffalla.

[9] Robyn M. Dawes. The robust beauty of improper linear models in decision making. *American Psychologist*, 34:571–582, 1979. doi:10.1037/0003-066X.34.7.571.

[10] Rachna Dhamija, J. D. Tygar, and Marti Hearst. Why phishing works. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, CHI, page 581–590. Association for Computing Machinery, April 2006. doi:10.1145/1124772.1124861.

[11] Ivan Dokmanic, Reza Parhizkar, Juri Ranieri, and Martin Vetterli. Euclidean distance matrices: Essential theory, algorithms, and applications. *IEEE Signal Processing Magazine*, 32(6):12–30, 2015. doi:10.1109/MSP.2015.2398954.

[12] Florian M. Farke, Lennart Lorenz, Theodor Schnitzler, Philipp Markert, and Markus Dürmuth. "You still use the password after all" – exploring FIDO2 security keys in a small company. In *Sixteenth Symposium on Usable Privacy and Security (SOUPS)*, pages 19–35, August 2020. URL: https://www.usenix.org/conference/soups2020/presentation/farke.

[13] FIDO Alliance. FIDO Alliance design guidelines for consumer use cases of passkeys. https://www.passkeycentral.org/design-guidelines/, June 2023.

[14] FIDO Alliance. FIDO2 — passkeys: Passwordless authentication. https://fidoalliance.org/fido2, 2026. FIDO Alliance official page on FIDO2 and passkeys.

[15] FIDO Alliance. Passkeys — passwordless authentication. https://fidoalliance.org/passkeys/, 2026. FIDO Alliance official resource explaining passkeys and their benefits.

[16] Chris Fisher. UX best practices for passkeys: Understanding device-initiated authentication. https://www.authsignal.com/blog/articles/ux-best-practices-for-passkeys-understanding-device-initiated-authentication, 2025. Authsignal blog; last updated May 14, 2025.

[17] Anthony Gavazzi, Ryan Williams, Engin Kirda, Long Lu, Andre King, Andy Davis, and Tim Leek. A study of

Multi-Factor and Risk-Based authentication availability. In *32nd USENIX Security Symposium*, pages 2043–2060, August 2023. URL: https://www.usenix.org/conference/usenixsecurity23/presentation/gavazzi.

[18] Eva Gerlitz, Maximilian Häring, Charlotte Theresa Mädler, Matthew Smith, and Christian Tiefenau. Adventures in recovery land: Testing the account recovery of popular websites when the second factor is lost. In *Nineteenth Symposium on Usable Privacy and Security (SOUPS)*, pages 227–243, August 2023. URL: https://www.usenix.org/conference/soups2023/presentation/gerlitz-adventures.

[19] Google, Inc. Passkeys user interface design. https://developers.google.com/identity/passkeys/ux/user-interface-design, 2025. Google Developers documentation; last updated 16 September 2025.

[20] Google LLC. Passkeys — cross-device user authentication, 2025. Overview of Passkeys on Google Developers, including usage, UX guidance, and cross-device authentication support. URL: https://developers.google.com/identity/passkeys.

[21] Nicolas Huaman, Sabrina Klivan, Marten Oltrogge, Yasemin Acar, and Sascha Fahl. They would do better if they worked together: The case of interaction problems between password managers and websites. In *IEEE Symposium on Security and Privacy (SP)*, pages 1367–1381, 2021. doi:10.1109/SP40001.2021.00094.

[22] Blake Ives, Kenneth R. Walsh, and Helmut Schneider. The domino effect of password reuse. *Communications of the ACM*, 47(4):75–78, April 2004. doi:10.1145/975817.975820.

[23] A.K. Jain, A. Ross, and S. Prabhakar. An introduction to biometric recognition. *IEEE Transactions on Circuits and Systems for Video Technology*, 14(1):4–20, 2004. doi:10.1109/TCSVT.2003.818349.

[24] Markus Keil, Philipp Markert, and Markus Dürmuth. "It's just a lot of prerequisites": A user perception and usability analysis of the German ID card as a FIDO2 authenticator. In *Proceedings of the European Symposium on Usable Security*, EuroUSEC, page 172–188. Association for Computing Machinery, 2022. doi:10.1145/3549015.3554208.

[25] Michal Kepkowski, Maciej Machulak, Ian Wood, and Dali Kaafar. Challenges with passwordless FIDO2 in an enterprise setting: A usability study. In *IEEE Secure Development Conference (SecDev)*, pages 37–48, 2023. doi:10.1109/SecDev56634.2023.00017.

[26] Sabrina Klivan, Sandra Höltervennhoff, Nicolas Huaman, Alexander Krause, Lucy Simko, Yasemin Acar, and Sascha Fahl. "We've disabled MFA for you": An evaluation of the security and usability of multi-factor authentication recovery deployments. In *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security*, CCS, page 3138–3152. Association for Computing Machinery, 2023. doi:10.1145/3576915.3623180.

[27] Leona Lassak, Annika Hildebrandt, Maximilian Golla, and Blase Ur. "It's stored, hopefully, on an encrypted server": Mitigating users' misconceptions about FIDO2 biometric WebAuthn. In *30th USENIX Security Symposium*, pages 91–108, August 2021. URL: https://www.usenix.org/conference/usenixsecurity21/presentation/lassak.

[28] Leona Lassak, Nicklas Lindemann, and Marvin Kowalewski. From TOTPs to security keys: Studying the reality of passwordless FIDO2 authentication with PIN and biometrics in a corporate environment. In *Twenty-First Symposium on Usable Privacy and Security (SOUPS)*, pages 371–389, August 2025. URL: https://www.usenix.org/conference/soups2025/presentation/lassak.

[29] Leona Lassak, Elleen Pan, Blase Ur, and Maximilian Golla. Why aren't we using passkeys? Obstacles companies face deploying FIDO2 passwordless authentication. In *33rd USENIX Security Symposium*, pages 7231–7248, August 2024. URL: https://www.usenix.org/conference/usenixsecurity24/presentation/lassak.

[30] Jonathan Lazar, Jinjuan Heidi Feng, and Harry Hochheiser. *Research Methods in Human-Computer Interaction*. Morgan Kaufmann, 2nd edition, 2017.

[31] Jeongho Lee, Hyoung-Kee Choi, Jin Hee Yoon, and Seongjune Kim. An empirical analysis of incorrect account remediation in the case of broken authentication. *IEEE Access*, 11:141610–141627, 2023. doi:10.1109/ACCESS.2023.3343411.

[32] Sanam Ghorbani Lyastani, Michael Backes, and Sven Bugiel. A systematic study of the consistency of two-factor authentication user journeys on top-ranked websites. In *Network and Distributed System Security (NDSS) Symposium*, pages 1–18, March 2023. doi:10.14722/ndss.2023.23362.

[33] Sanam Ghorbani Lyastani, Michael Schilling, Michaela Neumayr, Michael Backes, and Sven Bugiel. Is FIDO2 the kingslayer of user authentication? A comparative usability study of FIDO2 passwordless authentication.

In *IEEE Symposium on Security and Privacy (SP)*, pages 268–285, 2020. doi:10.1109/SP40000.2020.00047.

[34] Ahmed Tanvir Mahdad, Mohammed Jubur, and Nitesh Saxena. Breaching security keys without root: FIDO2 deception attacks via overlays exploiting limited display authenticators. In *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security*, CCS, page 1686–1700. Association for Computing Machinery, 2024. doi:10.1145/3658644.3690286.

[35] Microsoft Corporation. Design guidelines for passkeys. https://learn.microsoft.com/en-us/windows/apps/develop/security/design, 2024. Microsoft Learn documentation; Windows app security and passkey design guidance; last updated 9 September 2024.

[36] Mozilla Developer Network. PublicKeyCredentialRequestOptions, 2025. MDN Web Docs. Last modified April 14, 2025. URL: https://developer.mozilla.org/en-US/docs/Web/API/PublicKeyCredentialRequestOptions.

[37] Florian Nawrath. Quantitative Analysis of FIDO2 Client Support. In *Who Are You?! Adventures in Authentication Workshop*, WAY, pages 1–5, August 2021. URL: https://wayworkshop.org/2021/papers/way2021-nawrath.html.

[38] Nielsen Norman Group. Maintain consistency and adhere to standards. https://www.nngroup.com/articles/consistency-and-standards/, 2021. NN/g article on heuristic evaluation principle "Consistency and Standards"; published January 10, 2021.

[39] L. O'Gorman. Comparing passwords, tokens, and biometrics for user authentication. *Proceedings of the IEEE*, 91(12):2021–2040, 2003. doi:10.1109/JPROC.2003.819611.

[40] Okta, Inc. Passkeys primer: How to improve user experience and prevent account takeovers with phishing-resistant FIDO authentication, 2023. Whitepaper providing a primer on passkeys and their implementation; downloaded from Okta website. URL: https://www.okta.com/sites/default/files/2023-09/PasskeysPrimer-Whitepaper-082523-Final.pdf.

[41] Passkey Central. Design guidelines. Overview of Passkey Central's design guidelines for passkey UI/UX patterns and best practices. URL: https://www.passkeycentral.org/design-guidelines/.

[42] Passkey Central. How passkeys work. Explains how passkeys use public-key cryptography to provide user-friendly, phishing-resistant authentication. URL: https://www.passkeycentral.org/introduction-to-passkeys/how-passkeys-work.

[43] Nils Quermann, Marian Harbach, and Markus Dürmuth. The state of user authentication in the wild. In *Who Are You?! Adventures in Authentication Workshop*, WAY, 2018. URL: https://wayworkshop.org/2018/papers/way2018-quermann.pdf.

[44] N. K. Ratha, J. H. Connell, and R. M. Bolle. Enhancing security and privacy in biometrics-based authentication systems. *IBM Systems Journal*, 40(3):614–634, 2001. doi:10.1147/sj.403.0614.

[45] Maurice Roux. A comparative study of divisive and agglomerative hierarchical clustering algorithms. *Journal of Classification*, 35(2):345–366, 7 2018. doi:10.1007/s00357-018-9259-9.

[46] Florian Schaub, Marcel Walch, Bastian Könings, and Michael Weber. Exploring the design space of graphical passwords on smartphones. In *Ninth Symposium on Usable Privacy and Security (SOUPS)*, SOUPS, 2013. doi:10.1145/2501604.2501615.

[47] Statista GmbH. Market share held by operating systems worldwide since 2009. Statista statistics showing the worldwide market share of leading operating systems across devices since 2009. URL: https://www.statista.com/statistics/268237/global-market-share-held-by-operating-systems-since-2009/.

[48] Statista GmbH. Market share of internet browsers worldwide since 2009. Statista statistic showing the global market share trends of major internet browsers (e.g., Chrome, Safari, Edge, Firefox) from 2009 onward. URL: https://www.statista.com/statistics/268254/market-share-of-internet-browsers-worldwide-since-2009/.

[49] David Temoshok, Diana Proud-Madruga, Yee-Yin Choong, Ryan Galluzzo, Sarbari Gupta, Connie LaSalle, Naomi Lefkovitz, and Andrew Regenscheid. NIST SP 800-63-4: Digital identity guidelines. July 2025. doi:10.6028/NIST.SP.800-63-4.

[50] Ignacio Velásquez, Angélica Caro, and Alfonso Rodríguez. Kontun: A framework for recommendation of authentication schemes and methods. *Information and Software Technology*, 96:27–37, 2018. URL: https://www.sciencedirect.com/science/article/pii/S0950584917301714, doi:10.1016/j.infsof.2017.11.004.

[51] Stephan Wiefling, Luigi Lo Iacono, and Markus Dürmuth. Is this really you? An empirical study on risk-based authentication applied in the wild. In *ICT Systems Security and Privacy Protection*, pages 134–148. Springer International Publishing, 2019. doi:10.1007/978-3-030-22312-0_10.

[52] Leon Würsching, Florentin Putz, Steffen Haesler, and Matthias Hollick. FIDO2 the rescue? Platform vs. roaming authentication on smartphones. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, CHI, pages 1–16, April 2023. doi:10.1145/3544548.3580993.

[53] Verena Zimmermann, Nina Gerber, Marius Kleboth, Alexandra von Preuschen, Konstantin Schmidt, and Peter Mayer. The quest to replace passwords revisited - rating authentication schemes. In *Proceedings of the 12th International Symposium on Human Aspects of Information Security and Assurance (HAISA)*, pages 38–48, 2018. doi:10.5445/IR/1000085729.

A Comparison Factors Codebook

Table 4 presents each FIDO passkey design guideline categorized by design pattern. It also illustrates how we decomposed each guideline into descriptive elements used to derive our comparison factors. Table 5 outlines the process of identifying additional factors through open and axial coding.

B Examples From Websites

Figure 4 shows an example interface from the `adobe.com` login page that implements the Sign-In-UI (F17) and Fallback-Authentication (F19) factors. Figure 5 shows an interface screen from `authsignal.com` that implements both the Deletion-Confirmation (F28) and Authenticator-Cleanup (F25) factors. The interface prompts the user to confirm their intent to delete the passkey from the website and instructs them to also remove it from their Windows Hello passkey provider.

C Hierarchical Clustering

We used agglomerative hierarchical clustering with the average-linkage method to group websites by their passkey implementation practices. This process successively merged websites with similar implementations, resulting in 3 clusters, as shown in the dendrogram in Figure 6.

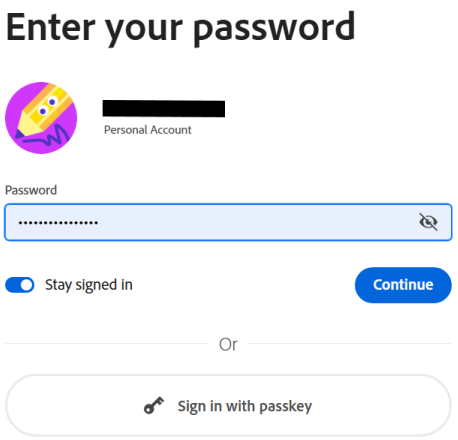


Figure 4: Sign-in page on `adobe.com`

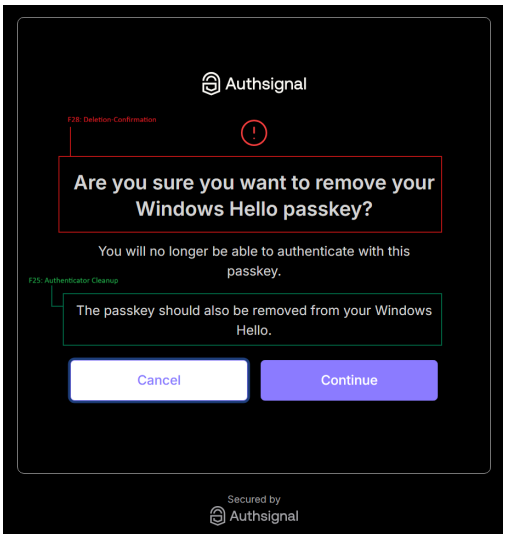


Figure 5: Deletion-Confirmation and Authenticator-Cleanup factors from `authsignal.com`

D Euclidean Distance

Figure 7 shows a heatmap of Euclidean distances among 111 websites based on their passkey implementation practices. Darker cells indicate higher similarity, corresponding to lower Euclidean distance, while lighter cells indicate greater dissimilarity. The heatmap reveals three natural groups of websites with similar passkey implementation practices, marked by colored bars along the axes. It also suggests that most websites form a cohesive main cluster of Standard Implementers with similar passkey implementation practices.

Table 4: Guidelines organized by design pattern category and deductive-coding derived factors

	Guideline	Description	Factor
Required	Create, View, and Manage Passkeys in Account Settings	Allow people to create passkeys in account settings	Settings-Setup (F08)
		Include a passkey hero prompt	Promotion (F01)
		Allow people to see each passkey as a card. One card per passkey	Management-UI (F23)
		Place the passkey UI above passwords in the interface	Priority (F02)
		Offer the ability to remove passkeys	Delete-Option (F24)
	Sign in with a Passkey	Enable autofill by adding autocomplete="webauthn" to the username input field	Autofill-Feature (F18)
		Support graceful fallback to other sign-in methods.	Fallback-Authentication (F19)
		Offer a dedicated sign in with a passkey button or link	Sign-In-UI (F17)
Optional	Create Passkey After Account Recovery Due To Forgotten Password	Add create passkey after account recovery due to forgotten password	Recovery-Setup (F11)
	Cross-Device Sign-In	Add cross-device setup	Cross-Device-Setup (F10)
		Add cross-device sign-in	Cross-Device-Usage (F20)
	Deprecate SMS OTP	Add deprecate SMS OTP	OTP-Setup (F07)
	New Account Creation with a Passkey	Add new account creation with a passkey	Registration-Setup (F06)
	Passkey Management UI: Best Practices for Combining All Passkey Types	Add management of different passkey types	Multicreation (F12)
	Remove Passkeys from Service Provider Account Settings	Confirm with the user before proceeding with passkey removal. When deleting the last passkey, inform user of the fallback authentication method. Include the option to cancel passkey removal.	Deletion-Confirmation (F28)
		Inform people to remove their passkeys from their credential managers after removal	Authenticator-Cleanup (F25)
		Confirm passkey removal with a success message	Deletion-Notification (F27)
	Use Passkeys Created on Websites to Sign In to Apps (and Apps to Websites)	Allow passkeys created on websites to sign in to apps (and vice-versa)	Mobile-App-Usage (F22)
	Create Passkey After Account Recovery Due To Forgotten Password	If people create a new password, allow them to create a passkey too	Recovery-Setup (F11)
Security-Key	Awareness of Security Keys as an Authentication Option	Add awareness of security keys	HSK-Awareness (F03)
	Enroll Passkeys on Security Keys	Add enroll passkeys on security keys	HSK-Setup (F09)
	Manage Passkeys on Security Keys. Sign in with Passkeys on Security Keys	Add manage passkeys on security keys, Add sign in with security keys as a second factor	HSK-Usage (F21)

Table 5: Open and axial coding process of deriving the additional pattern factors

	Open coding	Axial coding	Factor
Additional	Update passkeys, Rename passkeys, Edit passkeys	Offer the ability to update passkeys	Update-Option (F15)
	Identity proofing for passkeys setup, Reauthenticating user to setup passkeys, Require password to setup passkeys, Generate a one-time code to setup passkeys	Require identity verification prior to setting up a passkey	Setup-Authentication (F13)
	Identity proofing for passkeys deletion, Reauthenticating user to delete passkeys, Require password to delete passkeys, Generate a one-time code to delete passkeys	Require identity verification prior to deleting a passkey	Deletion-Authentication (F26)
	Follow approved nomenclature for passkeys, Use FIDO approved icons, Use FIDO approved naming	Use FIDO approved nomenclature, logos, and icons.	Naming-Conventions (F05)
	Passkey education, Include documentation about passkeys, Interfaces for educating users about passkeys	Provide educational content about the use of passkeys.	Educational-Resources (F04)

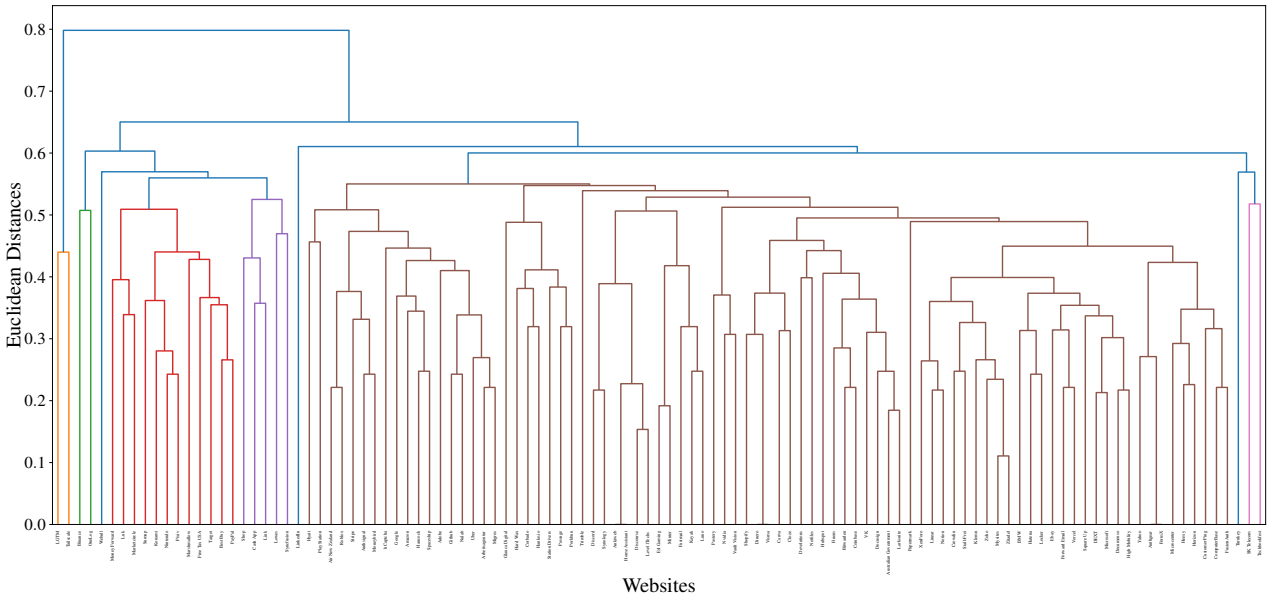


Figure 6: Dendrogram of Hierarchical Clustering

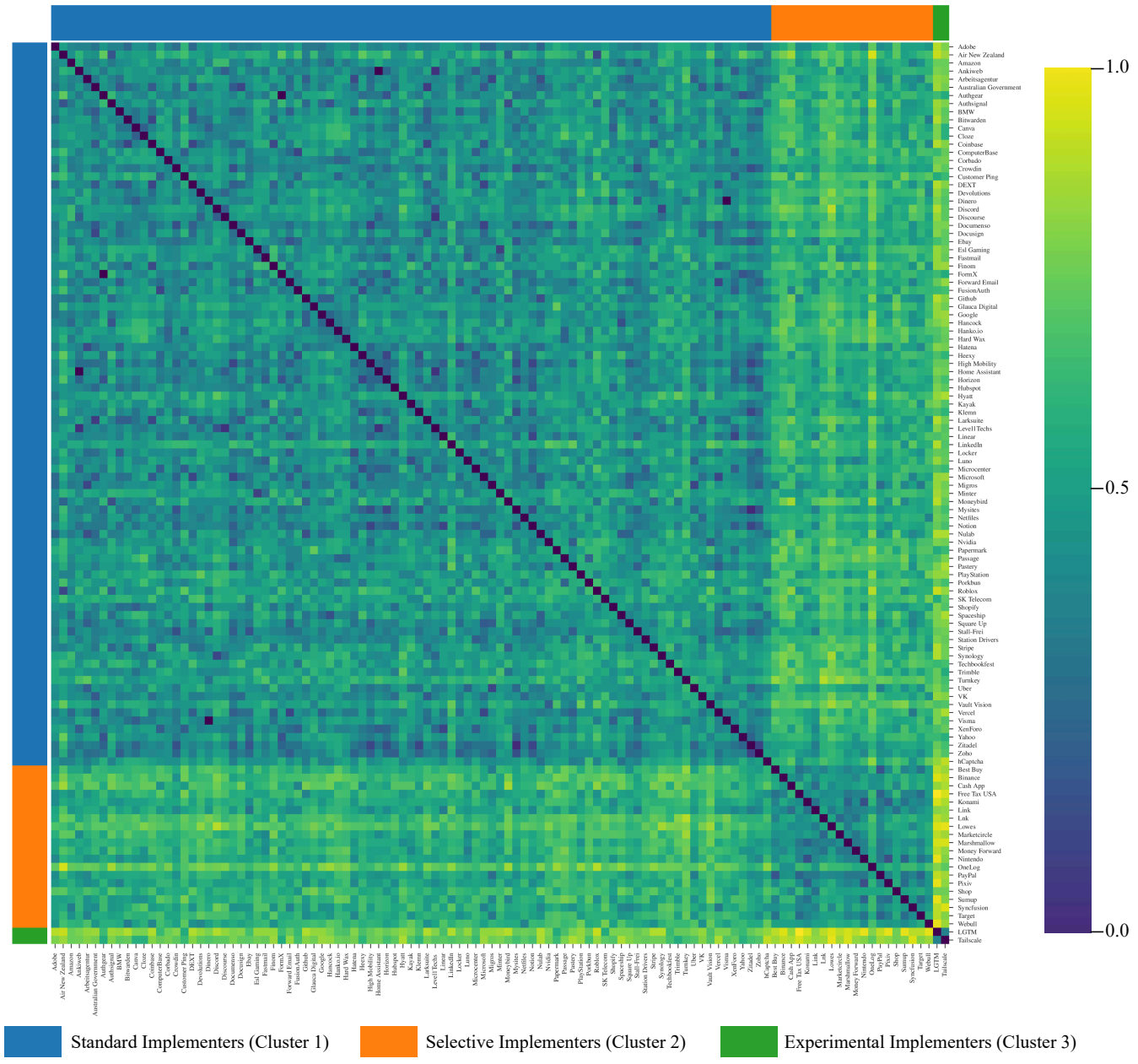


Figure 7: Heatmap of pairwise Euclidean distances